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Programmable AUTOMATIC BELL SYSTEM

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All schools and colleges follow a timetable for their classes, providing different time slots for different subjects. In order to follow the planned timetable, a person is assigned the job of turning on and off an electric bell manually at certain time intervals. Here we describe an automatic bell system that eliminates the need for human intervention. It is designed using simple digital electronics and is easy to construct. For clock and digital timer programming, a programmable time switch (Frontier TM-619) is used.

Circuit and working

Circuit diagram of the programmable automatic bell system is shown in Fig. 1. It is built around a step-down transformer (X1), 5V voltage regulator IC 7805 (IC1), three 74121 monostable ICs (IC2, IC3 and IC5), quad AND gate IC 7408 (IC4), six 1N4007 diodes (D1 through D6), 230V AC bell, digital timer programmable time switch (TM-619) and a few other components.

The working of the circuit can be understood easily by referring to the truth table of IC 74121. Truth table is given in the datasheet, which is easily available on the Internet. Else, you may refer this month's EFY DVD.

Fig. 1 shows how to use inputs of IC2 and IC3 for the required modes of operation, that is, falling-edge-triggered or rising-edge-triggered monostable multivibrator. Here, we use input A2 for falling-edge trigger of IC3, and input B for rising-edge trigger of IC2.

When we use input B for rising-edge trigger, A1, A2 or both these inputs of IC2 should be low, that is, connected to ground. Here, for convenience, both A1 and A2 are connected to ground. Similarly, for use of A2 as falling-edge trigger of IC3, pins A1 and B of IC3 should be high, that is, connected to 5V. Here IC2 is used as rising-edge-triggered, one-shot monostable multivibrator, whose time period is determined by the combination of resistor R1 and capacitor C4 as follows:

$$t = 0.69RC.$$

Here, value of R1 is taken as 10-kilo-ohm and C4 as 2.2µF for a pulse width of 1.5 milliseconds. Similarly, IC3 is used as a falling-edge-triggered one-shot monostable multivibrator, whose time period is determined by the combination of resistor R3 and capacitor C7. Here again, resistor and capacitor values are chosen for pulse duration of 1.5 milliseconds.

$\overline{Q}$ output of IC2 and IC3 are connected to input pins 2 and 1 of IC4, respectively. In normal condition, when there is no trigger to IC2 and IC3, $\overline{Q}$ outputs at pin 1 of

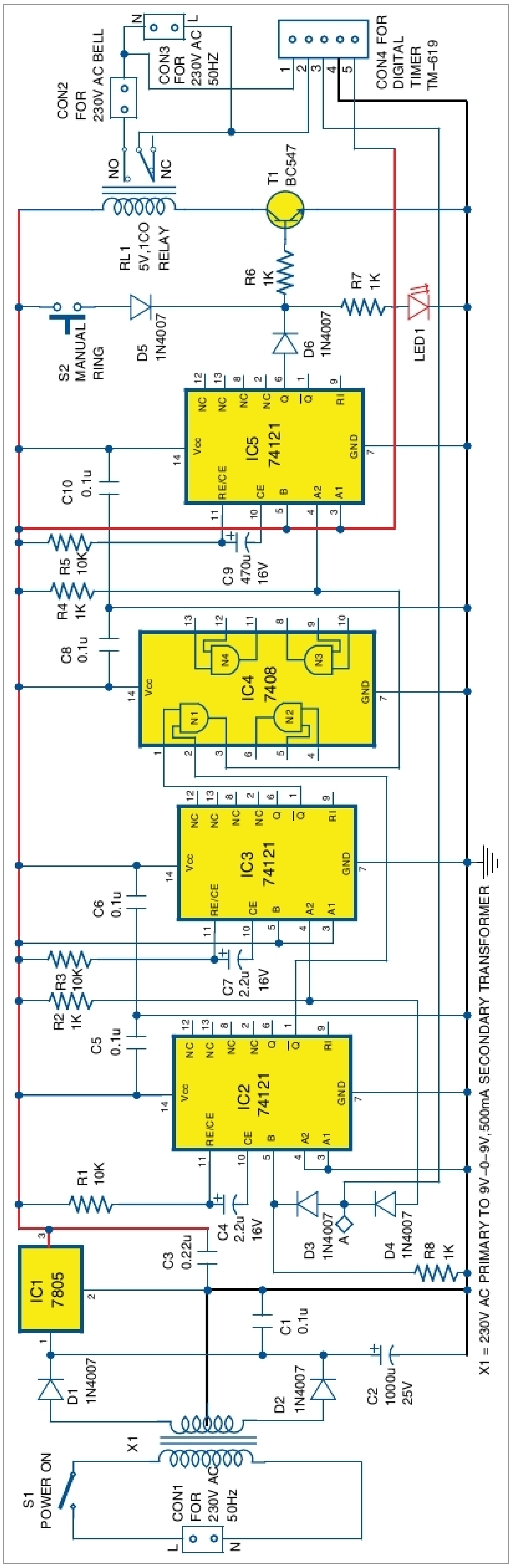

Fig. 1: Circuit diagram of programmable automatic bell system